Chapter 1. Introduction and Causal Models

Pearl's causal hierarchy, Simpson's paradox, structural causal models, causal graphs, d-separation

Contents

1.1 Why causal inference **2**
1.1.1 A brief review of machine learning . . . 2
1.1.1.1 Why machine learning? . . . 2
1.1.1.2 The classical learning paradigm . . . 2
1.1.1.3 A probabilistic view . . . 2
1.1.1.4 Reinforcement learning: intervention instead of observation . . . 2
1.1.2 Conceptual foundation: the ladder of causation . . . 3
1.1.2.1 Inference about worlds that are not realized . . . 3
1.1.2.2 Pearl's Causal Hierarchy . . . 3
1.1.2.3 Two motivations . . . 3
1.1.3 Correlation versus causation . . . 4
1.1.4 Example 1: Simpson's paradox in a 2×2 table . . . 4
1.1.4.1 Would the result be different with a different story? . . . 5
1.1.5 Example 2: a "trivial" data analysis . . . 5
1.1.6 What is going on here? . . . 5

1.2 Causal models and causal graphs **6**
1.2.1 Reviews: notation, probability, algebra . . . 6
1.2.1.1 Notation . . . 6
1.2.1.2 Probability . . . 6
1.2.1.3 Sums of products . . . 6
1.2.2 Why data alone is not enough: a prima facie example . . . 7
1.2.2.1 The story . . . 7
1.2.2.2 Current protocol . . . 7
1.2.2.3 Hypothetical protocols . . . 7
1.2.2.4 What can a learning system see? . . . 7
1.2.2.5 From statistical to causal analysis . . . 8
1.2.2.6 Modeling reality in the example . . . 8
1.2.3 Structural causal models . . . 8
1.2.3.1 Property 1: an SCM induces the observational distribution . . . 8
1.2.3.2 Property 2: an SCM induces a causal diagram . . . 9
1.2.3.3 Graph terminology . . . 10
1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy . . . 10
1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$ . . . 10
1.2.5 Conditional independence and the Markov condition . . . 11
1.2.5.1 Conditional independence . . . 11
1.2.5.2 Local Markov condition . . . 11
1.2.5.3 Markovian models and the Bayesian factorization . . . 11
1.2.6 Reading independences from the graph: d-separation . . . 11
1.2.6.1 Chains . . . 11
1.2.6.2 Common causes (forks) . . . 12
1.2.6.3 Common effects (colliders) . . . 12
1.2.6.4 Triplet summary . . . 12
1.2.6.5 d-separation . . . 12
1.2.7 Implementation: graph algorithms in Python . . . 13
1.2.7.1 Topological order . . . 13
1.2.7.2 d-separation via reachability (Markovian graphs) . . . 14

1.1 Why causal inference

Motivation. Classical ML and statistics learn aspects of a *fixed* data-generating reality, which is not enough for questions about actions and unrealized worlds.

1.1.1 A brief review of machine learning

Motivation. What standard learning paradigms estimate shows what they *cannot* estimate.

1.1.1.1 Why machine learning?

Intelligent behavior needs a model of the environment; writing it down is often infeasible, so it is *learned from data through algorithms*.

Learning modifies an agent's decision mechanism to improve performance and adapt to unexpected situations. Three questions recur:

- How can the learning process be characterized?
- How should prior knowledge be traded off against performance?
- How robust and generalizable is the resulting system?

Applications: generative AI, AI for science (protein folding, drug discovery), medicine, sustainability, vision, autonomous systems and security.

1.1.1.2 The classical learning paradigm

An unknown true model M generates feature vectors x; the target is a function $f_M(x)$. From $\{x^{(i)}\}_{i=1}^N$ one selects $h \in \mathcal{H}$ with $h(x) \approx f_M(x)$; the challenge is generalization.

Supervised learning maps $\mathcal{X}$ to $\mathcal{Y}$ from pairs $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$, mimicking an oracle $f : \mathcal{X} \to \mathcal{Y}$ (trees, k-NN, Bayesian networks, CNNs, Transformers, XGBoost).

Unsupervised learning sees only $x \in \mathcal{X}$ and seeks structure: clustering, dimensionality reduction, density modeling of $P(x)$ (e.g. a phylogenetic tree).

1.1.1.3 A probabilistic view

Both estimate features of one fixed joint distribution:

- supervised learning infers $P(y \mid x)$ (a network's output activation, a tree leaf);
- unsupervised learning infers properties of $P(x)$.

We learn to imitate aspects of a static reality; $P(x, y)$ says nothing about changing the mechanism producing x.

1.1.1.4 Reinforcement learning: intervention instead of observation

In RL the agent *intervenes* in the environment and learns from its responses (A/B testing, robotics), facing the exploration/exploitation trade-off. Missing: reasoning about actions from data *not* generated by our own experiments, and about events that did not happen.

1.1.2 Conceptual foundation: the ladder of causation

Motivation. Observing, intervening and imagining structure the whole course.

Layer 1. Observation (sensors): observe the world and infer the "next" stage.

Layer 2. Agency: intervene in the world and change it.

Layer 3. Retrospection: reflect on what we did (and did not do) and infer new behaviors.

This stratification is formalized as **Pearl's Causal Hierarchy (PCH)**.

1.1.2.1 Inference about worlds that are not realized

The classical pipeline is

$$\text{Real world / nature} \xrightarrow{\text{data}} \text{AI/ML, statistics (inference)} \longrightarrow \text{new knowledge.}$$

An unrealized *alternative reality* has no data, so the only route is

$$\text{knowledge} \longrightarrow \text{causal model} \xrightarrow{\text{causal inference}} \text{new knowledge about the alternative reality,}$$

with mechanism assumptions transporting information to unobserved worlds.

1.1.2.2 Pearl's Causal Hierarchy

Level (symbol)	Activity	Question	Examples / typical models
1. Association	Seeing	What is?	What does a symptom tell us about a disease? How would seeing X change my belief in Y?
$P(y \mid x)$			(Un)supervised ML: deep nets, Bayes nets, hierarchical models, decision trees.
2. Intervention	Doing	What if?	What if I take aspirin, will my headache be cured? What if I do X?
$P(y \mid \text{do}(x), c)$			Reinforcement learning, causal Bayesian networks, MDPs, POMDPs.
3. Counterfactual	Imagining,	Why?	Was it the aspirin that stopped my headache? What if I had acted differently?
$P(y_x \mid x', y')$	retrospection		Structural causal models.

Causal inference = cross-layer inference.

- *Layer 1 data, Layer 2 query:* will performance improve if class starts earlier, from observational records?
- *Layer 2 data, Layer 2 query:* predict a new experiment from previous ones (transportability).
- *Layers 1–2 data, Layer 3 query:* would I have won with door #2 instead of door #1?

Lower layers generally *underdetermine* higher ones (proved later).

1.1.2.3 Two motivations

- **Scientific:** understanding nature and humans with data, logic, mathematics and the scientific method.
- **Pragmatic** (economics, medicine, engineering): intelligent systems and policy design; policies are interventions (Layer 2).

1.1.3 Correlation versus causation

Motivation. Strong association is compatible with no causal relation at all.

- Active Facebook users and 10-year Greek bond yields rise together, 2005–2011 (BusinessWeek, Dec. 2011).
- Lemons imported from Mexico and US highway fatality rate, 1996–2000: $R^2 = 0.97$ (negative slope).
- Organic food sales and diagnosed autism, 1998–2007: $r = 0.9971$ ($p < 0.0001$).
- Maine divorce rate and per-capita margarine consumption (2000–2009).

These reflect common trends or coincidence (cf. xkcd no. 552). The real question: *if correlation does not imply causation, what does?* Answer: correlation *plus* causal assumptions encoded in a model.

1.1.4 Example 1: Simpson's paradox in a 2×2 table

Motivation. The same data support opposite conclusions under different causal stories.

$X/\neg X$: drug/no drug; $Y/\neg Y$: recovered/not; $F/\neg F$: female/male (Simpson 1951; Pearl 2000, Ch. 6).

All	Y	$\neg Y$	total	rate
drug (X)	20	20	40	50%
no drug ($\neg X$)	16	24	40	40%
Σ	36	44	80	

Male ($\neg F$)	Y	$\neg Y$	total	rate
drug	18	12	30	60%
no drug	7	3	10	70%
Σ	25	15	40	

Female (F)	Y	$\neg Y$	total	rate
drug	2	8	10	20%
no drug	9	21	30	30%
Σ	11	29	40	

Within each sex the drug is worse,

$$P(Y \mid F, X) = 0.2 < 0.3 = P(Y \mid F, \neg X),$$
$$P(Y \mid \neg F, X) = 0.6 < 0.7 = P(Y \mid \neg F, \neg X),$$

yet overall better: $P(Y \mid X) = 0.5 > 0.4 = P(Y \mid \neg X)$.

Why this is arithmetically possible. By total probability,

$$P(Y \mid X) = P(Y \mid X, \neg F)P(\neg F \mid X) + P(Y \mid X, F)P(F \mid X),$$

with $P(F \mid X) = 10/40 = 0.25$, $P(F \mid \neg X) = 30/40 = 0.75$:

$$P(Y \mid X) = 0.75 \cdot 0.6 + 0.25 \cdot 0.2 = 0.50,$$
$$P(Y \mid \neg X) = 0.25 \cdot 0.7 + 0.75 \cdot 0.3 = 0.40.$$

Different weights: men, who recover more often anyway, are over-represented among drug takers.

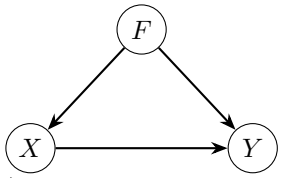
Which table should the physician consult? If sex affects both treatment and recovery, weight the strata by the *population* $P(F) = P(\neg F) = 1/2$:

$$\sum_f P(Y \mid X, f)P(f) = \tfrac{1}{2}(0.6) + \tfrac{1}{2}(0.2) = 0.40,$$
$$\sum_f P(Y \mid \neg X, f)P(f) = \tfrac{1}{2}(0.7) + \tfrac{1}{2}(0.3) = 0.50,$$

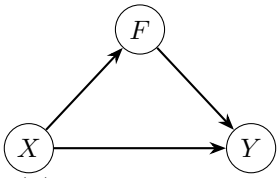
so the drug lowers recovery by 10 points (back-door adjustment, Chapter 2).

1.1.4.1 Would the result be different with a different story?

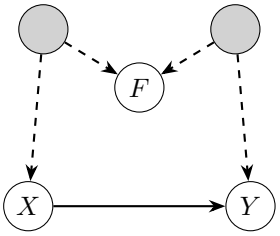
Same numbers, three graphs:



(a) F is a common cause



(b) F is a mediator



(c) F is a collider of latents

- **(a)** $F \to X$, $F \to Y$, $X \to Y$. F confounds: use the stratified tables; the drug is harmful.
- **(b)** $X \to F \to Y$, $X \to Y$. If the third variable is changed by the drug (e.g. blood pressure), stratifying blocks part of the effect; the aggregate table gives the total effect: beneficial.
- **(c)** $X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$, U_1, U_2 latent. Conditioning on the collider F *creates* spurious association; the aggregate table is correct.

Identical data, different answers: the decision needs the causal structure, outside the data.

1.1.5 Example 2: a "trivial" data analysis

Motivation. A perfect fit does not reveal the direction or meaning of a dependence.

Firefighters (X)	Size of fire (Y)
10	100
40	400
60	600
200	2000

$Y = 10X$ exactly, so $\beta = 10$, $\beta_0 = 0$. Read causally, each firefighter adds ten units of fire; in fact $Y \to X$. *What is the meaning of β?* A property of the joint distribution; whether it measures the effect of changing X depends on the mechanism.

1.1.6 What is going on here?

Both examples ask for the effect of an action, which is not in the joint distribution. This needs a formal language for causal assumptions (i), semantics linking assumptions, data and queries (ii), and inference rules for computability (iii). Chapters 1–3 build it: SCMs, causal graphs, identification, back-door and adjustment criteria, do-calculus, algorithmic identification.

1.2 Causal models and causal graphs

Motivation. Which object can serve as an "oracle" for causal questions, and what does it imply about observed data?

1.2.1 Reviews: notation, probability, algebra

Motivation. Notation and elementary manipulations used throughout.

1.2.1.1 Notation

- Uppercase Z: random variable with state space $\mathfrak{X}_Z$; lowercase $z \in \mathfrak{X}_Z$: a value.
- Bold $\mathbf{V}$, $\mathbf{v}$: sets of variables/values; calligraphic $\mathcal{G}$, $\mathcal{M}$: graphs, models.
- $\forall$, $\exists$; $\cup, \cap$; $\subseteq, \subset$ ($\subset$ is ambiguous); $\vee$, $\wedge$, $\oplus$ (xor); $\sum$, $\prod$; $\langle \cdots \rangle$ tuples, $\{\cdots\}$ sets, $\mid$ "given".
- "iff" = if and only if. $p \Rightarrow q$: p is *sufficient* for q; equivalently $\neg q \Rightarrow \neg p$: q is *necessary* for p.

1.2.1.2 Probability

For $P(A, B, C, D, E, F)$ write $P(a, b)$ for $P(A = a, B = b)$; variables are discrete WLOG.

Marginalization.

$$\sum_{c,e} P(a, b, c, d, e, f) = \sum_{(c,e) \in \mathfrak{X}_C \times \mathfrak{X}_E} P(a, b, c, d, e, f) = P(a, b, d, f).$$

Conditioning.

$$P(a, b, c \mid d, e) = \frac{P(a, b, c, d, e)}{P(d, e)} = \frac{\sum_f P(a, b, c, d, e, f)}{\sum_{a,b,c} \sum_f P(a, b, c, d, e, f)}.$$

Chain rule (in *any* order):

$$P(a, b, d, e) = P(a \mid b, d, e) P(b \mid d, e) P(d \mid e) P(e) = P(b \mid a, d, e) P(d \mid a, e) P(e \mid a) P(a).$$

Axioms. $0 \leq P(\cdot) \leq 1$, $\sum_{a,b,c} P(a, b, c) = 1$; often $P > 0$ so all conditionals exist.

1.2.1.3 Sums of products

Factors free of a summation index move outside the sum:

$$\sum_{a,c} f(a, b) g(b, c) h(c, d) = \sum_c h(c, d) g(b, c) \Big(\sum_a f(a, b) \Big) = \sum_a f(a, b) \Big(\sum_c g(b, c) h(c, d) \Big).$$

If $f(a, b) = P(a \mid b)$,

$$\sum_a P(a \mid b) \underbrace{\sum_c g(b, c) h(c, d)}_{\text{irrelevant to } a} = \underbrace{\Big(\sum_a P(a \mid b) \Big)}_{=1} \sum_c g(b, c) h(c, d) = \sum_c g(b, c) h(c, d),$$

the "sum out a leaf" step used constantly in identification proofs.

1.2.2 Why data alone is not enough: a prima facie example

Motivation. Every learner trained on observational data can recommend the wrong policy.

1.2.2.1 The story

A drug is *believed* to help against a contagious disease. Unknown to physicians:

- the rich always survive;
- carriers of a protective gene need no treatment, but if treated develop an allergy that is fatal under poor living conditions.

Wealth and gene are independent, $P(\text{rich}) = P(\text{poor}) = \frac{1}{2}$, and half the poor carry the gene:

$$\Pi = \text{rich} \ \cup \ \text{poor}_1 \ \cup \ \text{poor}_2,$$

$$\begin{cases} \text{rich}: & \text{alive anyway}, \\ \text{poor}_1 \ (\text{no gene}): & \text{die anyway}, \\ \text{poor}_2 \ (\text{gene}): & \text{die iff they take the drug}, \end{cases}$$

with $P(\text{poor}_1) = P(\text{poor}_2) = \frac{1}{4}$. (*Prima facie* = at first sight.)

1.2.2.2 Current protocol

The expensive drug goes to the rich only. All treated survive; among the untreated, poor_2 survive and poor_1 die:

$$P(\text{alive} \mid \text{drug}) = 1, \qquad P(\text{alive} \mid \neg\text{drug}) = 0.5.$$

Should the city give the drug to everyone for free?

1.2.2.3 Hypothetical protocols

Group (mass)	Current (drug iff rich)	Drug for all	Drug for none
rich (1/2)	alive	alive	alive
poor_1 (1/4)	dead	dead	dead
poor_2 (1/4)	alive	dead	alive
$P(\text{alive})$	3/4	1/2	3/4

Drug for all *reduces* survival from 3/4 to 1/2; the drug never helps, and the apparent benefit comes from treating only the rich.

1.2.2.4 What can a learning system see?

Rows (patient, R, D, A), e.g. (1, yes, yes, yes), (2, no, no, yes), (3, no, no, no), (4, yes, yes, yes),... At best any algorithm learns

R	D	A	$P(r, d, a)$
0	0	0	1/4
0	0	1	1/4
1	1	1	1/2
others			0

Unseen: $P(\text{gene})$, the other causes of wealth, and how the gene interacts with wealth, drug and reaction.

What is missing? In a second world the drug is inert, $A \leftarrow R \vee U_G$. Since $D = R$, it yields the same table, but drug for all gives 3/4. No algorithm using only $P(r, d, a)$ is right in both worlds; data must be supplemented by a model of how they are generated.

1.2.2.5 From statistical to causal analysis

- **Traditional statistics/ML:** data $\to P \to Q(P)$, e.g. $Q = P(B \mid A)$ for co-purchases.
- **Causal analysis:** data $\to P \xrightarrow{\text{change}} P' \to Q(P')$, e.g. sales if the price doubles, cancer if smoking is banned. P' is never observed; a formalism must represent both.
- **SCM paradigm:** data $\leftarrow P \leftarrow \mathcal{M} \to Q(\mathcal{M})$, $\mathcal{M}$ an invariant mechanism by which Nature assigns values. *P is a model of data; $\mathcal{M}$ is a model of reality.*

This refines the picture of Section 1.1 (with a grain of salt).

1.2.2.6 Modeling reality in the example

Endogenous $\mathbf{V}$: R (rich), D (drug), A (alive). Exogenous $\mathbf{U}$: U_G (gene), U_R (other causes of wealth). Mechanisms:

$$R \leftarrow U_R, \qquad D \leftarrow R, \qquad A \leftarrow R \vee (U_G \wedge \neg D),$$

$P(U_G = 1) = P(U_R = 1) = \frac{1}{2}$, $P(\mathbf{U}) = P(U_G)P(U_R)$. This fully specified model implies P and every P': the *structural causal model*.

1.2.3 Structural causal models

Motivation. Define the object representing reality and its three basic properties.

Definition (Structural causal model; Pearl 2000)

An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$ where

- $\mathbf{V} = \{V_1, \ldots, V_n\}$ are endogenous variables;
- $\mathbf{U}$ is a set of exogenous variables;
- $\mathcal{F} = \{f_1, \ldots, f_n\}$ are functions determining $\mathbf{V}$, $v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$, with $Pa_i \subseteq \mathbf{V} \setminus \{V_i\}$ and $\mathbf{U}_i \subseteq \mathbf{U}$;
- $P(\mathbf{U})$ is a distribution over $\mathbf{U}$.

"$\leftarrow$" marks an assignment, not a symmetric equation (axiomatizations: Galles and Pearl 1998, Halpern 1998). $\mathbf{U}$ carries all randomness; $\mathcal{F}$ maps a unit $\mathbf{u}$ to its observable profile.

1.2.3.1 Property 1: an SCM induces the observational distribution

$\mathcal{F} : \mathbf{u} \mapsto (v_1, \ldots, v_n)$, so random $\mathbf{U}$ induces $P(\mathbf{v})$, Layer 1 of the PCH; each unit is an instantiation $\mathbf{U} = \mathbf{u}$.

Definition (Induced distribution)

$\mathcal{M}$ defines $P^{\mathcal{M}}(\mathbf{V})$ by, for each $\mathbf{Y} \subseteq \mathbf{V}$,

$$P^{\mathcal{M}}(\mathbf{y}) = \sum_{\{\mathbf{u}\,:\,\mathbf{Y}(\mathbf{u})=\mathbf{y}\}} P(\mathbf{u}),$$

where $\mathbf{Y}(\mathbf{u})$, the *potential response*, is the value of $\mathbf{Y}$ obtained by solving $\mathcal{F}$ at $\mathbf{u}$.

Example 1.1

In the drug example:

$P(\mathbf{u})$	(U_R, U_G)	$\xrightarrow{\mathcal{F}}$ (R, D, A)	$P(\mathbf{v})$
1/4	(1,1)	(1,1,1)	1/2
1/4	(1,0)	(1,1,1)	
1/4	(0,1)	(0,0,1)	1/4
1/4	(0,0)	(0,0,0)	1/4

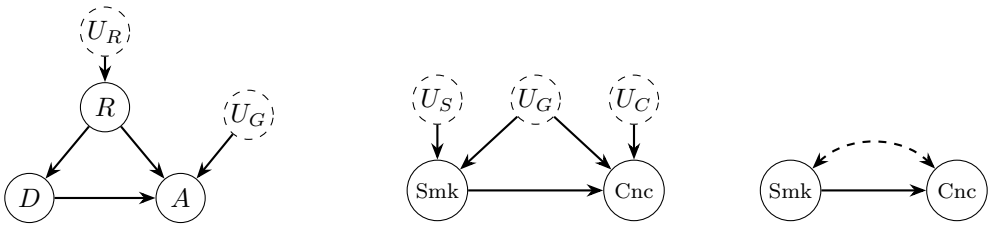
1.2.3.2 Property 2: an SCM induces a causal diagram

Definition (Causal diagram; Bareinboim et al. 2020, Def. 13)

Given $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$, the causal diagram $\mathcal{G}$ is constructed by

1. adding a vertex for every $V_i \in \mathbf{V}$;
2. adding a directed edge $V_j \to V_i$ whenever V_j is an argument of f_i;
3. adding a bidirected edge $V_j \leftrightarrow V_i$ whenever $\mathbf{U}_i$ and $\mathbf{U}_j$ are correlated or f_i, f_j share some $U \in \mathbf{U}$ as an argument.

Precisely: for every vertex set $\mathbf{W}$ without bidirected edges among them, $\{\mathbf{U}_W\}_{W \in \mathbf{W}}$ are jointly independent. With acyclic directed part (assumed throughout) $\mathcal{G}$ is an ADMG. E.g. $V_i \leftarrow f_i(A, B, U)$, $V_j \leftarrow f_j(C, U)$ give $A \to V_i \leftarrow B$, $C \to V_j$, $V_i \leftrightarrow V_j$.



drug example explicit $\mathbf{U}$ $\mathbf{U}$ implicit

Example 1.2 (Smoking and cancer)

$\mathbf{V} = \{\text{Smoking}, \text{Cancer}\}$, $\mathbf{U} = \{U_S, U_C, U_G\}$, Smoking $\leftarrow f_S(U_S, U_G)$, Cancer $\leftarrow f_C(\text{Smoking}, U_C, U_G)$; the shared U_G gives Smoking $\leftrightarrow$ Cancer.

Remark 1. A graph is compatible with infinitely many SCMs with the same function signatures and exogenous structure.

Remark 2. Researchers often know the *scope* of mechanisms but not their form; causal inference

asks what graph plus data imply.
Convention. Latents are implicit; a shared or correlated U is a dashed bidirected edge.

1.2.3.3 Graph terminology

$\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$; for $V \in \mathbf{V}$:

- parents $Pa(V) = \{W : W \to V \in \mathcal{G}\}$; children $Ch(V) = \{W : V \to W \in \mathcal{G}\}$;
- ancestors $An(V) = \{W : W \to \cdots \to V \in \mathcal{G}\}$; descendants $De(V) = \{W : V \to \cdots \to W \in \mathcal{G}\}$; both include V;
- siblings $\{W : W \leftrightarrow V \in \mathcal{G}\}$;
- a *path* joins distinct vertices by edges; a *directed path* from W to V has all arrowheads toward V.

1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy

$\mathcal{M}$ answers $P(y \mid x)$, $P(y \mid \mathrm{do}(x), c)$ and $P(y_x \mid x', y')$. Here Layer 1; Layers 2–3 in Chapter 2.

1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$

Motivation. Derive the factorization of the observational distribution implied by an SCM.

$\mathcal{M}$ induces

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{V} = \mathbf{v}, \mathbf{U} = \mathbf{u}) = \sum_{\mathbf{u}} P\big(\forall_{V_i \in \mathbf{V}}\, f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i,\ \mathbf{U} = \mathbf{u}\big),$$

$\mathbf{pa}_i, \mathbf{u}_i$ being the parents of V_i in $\mathcal{G}$. In the examples,

$$P(r, d, a) = \sum_{u_R, u_G} P(\underbrace{r, d, a}_{\text{endogenous}}, \underbrace{u_R, u_G}_{\text{exogenous}}),$$
$$P(s, c) = \sum_{u_S, u_G, u_C} P(s, c, u_S, u_G, u_C).$$

Topological order. $\mathcal{G}$ acyclic $\Rightarrow$ there is an order $\prec$ with $Pa_i \prec V_i$, e.g. $R \prec D \prec A$, $S \prec C$.

Factorization. By the chain rule along $\prec$,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}),$$

and since $Pa_i \cup \mathbf{U}_i \subseteq \{V_1, \ldots, V_{i-1}\} \cup \mathbf{U}$ determine V_i,

$$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P\big(f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}\big) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

Therefore

$$\boxed{P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).}$$

Examples:

$$P(r, d, a) = \sum_{u_R, u_G} P(u_R, u_G) P(r \mid u_R) P(d \mid r) P(a \mid r, d, u_G),$$
$$P(s, c) = \sum_{u_S, u_G, u_C} P(u_S, u_G, u_C) P(s \mid u_G, u_S) P(c \mid s, u_G, u_C),$$

with $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) \in \{0, 1\}$ (deterministic mechanisms).

1.2.5 Conditional independence and the Markov condition

Motivation. Translate the structure of $\mathcal{G}$ into testable constraints on $P(\mathbf{v})$.

1.2.5.1 Conditional independence

$$X \perp\!\!\!\perp Y \iff P(y \mid x) = P(y) \;\; \forall x, y \iff P(x, y) = P(x)P(y),$$
$$(X \perp\!\!\!\perp Y \mid Z) \iff P(x \mid z, y) = P(x \mid z) \quad \forall x, y, z \text{ with } P(y, z) > 0.$$

Intuition: *lack of functional dependence leads to probabilistic independence.*

1.2.5.2 Local Markov condition

$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ in any topological order, and the largest set that can precede V_i is its *non-descendants* Nd_i. Hence

$$P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = P(v_i \mid \mathbf{nd}_i, \mathbf{u}),$$

i.e.

$$V_i \perp\!\!\!\perp (Nd_i \setminus Pa_i), \; (\mathbf{U} \setminus \mathbf{U}_i) \;\mid\; Pa_i, \mathbf{U}_i.$$

1.2.5.3 Markovian models and the Bayesian factorization

If $U_1, \ldots, U_n$ are jointly independent (no bidirected edges), the model is *Markovian*:

$$\begin{aligned}
P(\mathbf{v}) &= \sum_{\mathbf{u}} P(\mathbf{u}) \prod_i P(v_i \mid \mathbf{pa}_i, u_i) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i) && (U_i \text{ independent}) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i \mid \mathbf{pa}_i) && (U_i \perp\!\!\!\perp Pa_i) \\
&= \sum_{\mathbf{u}} \prod_i P(v_i, u_i \mid \mathbf{pa}_i) \\
&= \prod_i \sum_{u_i} P(v_i, u_i \mid \mathbf{pa}_i) = \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i).
\end{aligned}$$

The sum distributes because the i-th factor depends only on u_i; $U_i \perp\!\!\!\perp Pa_i$ because Pa_i are functions of non-descendants' exogenous variables.

Summary. Markovian SCM $\Rightarrow P(\mathbf{v}) = \prod_i P(v_i \mid \mathbf{pa}_i) \Leftrightarrow$ local Markov $V_i \perp\!\!\!\perp Nd_i \setminus Pa_i \mid Pa_i$.

1.2.6 Reading independences from the graph: d-separation

Motivation. Three elementary structures link mechanisms to observable independences.

Binary variables, explicit mechanisms.

1.2.6.1 Chains

$X \to Z \to Y$: study habits $X \leftarrow U_X$, high Suneung score $Z \leftarrow X \vee \neg U_Z$ (U_Z: difficulty), admission $Y \leftarrow Z \wedge U_Y$. $X \perp\!\!\!\perp Y \mid Z$? **Yes**:

$$P(x, y \mid z) = \frac{P(x, y, z)}{P(z)} = \frac{P(y \mid z)P(z \mid x)P(x)}{P(z)} = \frac{P(x, z)}{P(z)} P(y \mid z) = P(x \mid z)P(y \mid z).$$

Observing Z blocks the chain (marginally X, Y are dependent).

1.2.6.2 Common causes (forks)

$X \leftarrow Z \rightarrow Y$: rain $Z \leftarrow U_Z$, traffic $X \leftarrow Z \vee \neg U_X$, umbrella $Y \leftarrow Z \wedge U_Y$. $X \perp\!\!\!\perp Y$? **No**: an umbrella makes rain, hence traffic, more likely. Given Z? **Yes**:

$$P(x, y \mid z) = \frac{P(x \mid z)P(y \mid z)P(z)}{P(z)} = P(x \mid z)P(y \mid z).$$

1.2.6.3 Common effects (colliders)

$X \rightarrow Z \leftarrow Y$: cold $X \leftarrow U_X$, holiday $Y \leftarrow U_Y$, attending class $Z \leftarrow \neg Y \wedge (\neg X \wedge U_Z)$. $X \perp\!\!\!\perp Y$? **Yes**:

$$P(x, y) = \sum_z P(x)P(y)P(z \mid x, y) = P(x)P(y)\sum_z P(z \mid x, y) = P(x)P(y).$$

Given Z? **No!** Absent ($Z = 0$) on a non-holiday ($Y = 0$) makes a cold more likely ("explaining away").

Given a descendant $W \leftarrow Z \wedge U_W$ (signing the sheet)? **No, again**: $W = 0$ makes $Z = 0$ likelier. *Watch out for descendants of colliders.*

1.2.6.4 Triplet summary

Triplet	Active (dependence flows) when	Inactive when
chain $X \rightarrow Z \rightarrow Y$	Z not observed	Z observed
fork $X \leftarrow Z \rightarrow Y$	Z not observed	Z observed
collider $X \rightarrow Z \leftarrow Y$	Z or a descendant of Z observed	neither Z nor its descendants observed

1.2.6.5 d-separation

Definition (d-separation)

To decide whether $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ (with $X, Y \notin \mathbf{Z}$):

1. look at every path between X and Y;
2. a path is *active* given $\mathbf{Z}$ if every consecutive triplet on it is active;
3. if any path is active, X and Y are d-connected (not guaranteed independent); otherwise they are d-separated, written $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$.

Trivially, if $X = Y$ or X and Y are adjacent, they are dependent.

Soundness: $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}} \Rightarrow X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in every distribution induced by an SCM compatible with $\mathcal{G}$ (global Markov property, equivalent to local for DAGs).

Example 1.3 (Sprinkler network)

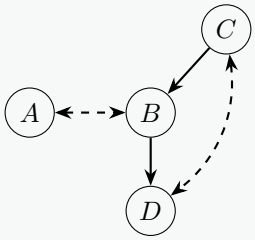
Season $\rightarrow$ Sprinkler, Season $\rightarrow$ Rain, Sprinkler $\rightarrow$ Wet, Rain $\rightarrow$ Wet, Wet $\rightarrow$ Slippery.

1. Wet $\perp\!\!\!\perp$ Sprinkler? **No**, adjacent.
2. Wet $\perp\!\!\!\perp$ Season $\mid$ Sprinkler? **No**: Season $\rightarrow$ Rain $\rightarrow$ Wet is open.
3. Rain $\perp\!\!\!\perp$ Slippery $\mid$ Wet? **Yes**: the only path is the chain through Wet.
4. Season $\perp\!\!\!\perp$ Wet $\mid$ Sprinkler, Rain? **Yes**: both directed paths are blocked.

5. Sprinkler $\perp\!\!\!\perp$ Rain | Season, Wet? **No**: Wet is an observed collider.

Example 1.4 (Food for thought [Supplement])

Graph: $A \leftrightarrow B$, $C \to B$, $B \to D$, $C \leftrightarrow D$.



- $A \perp\!\!\!\perp D$? **No**: B is an unobserved non-collider on $A \leftrightarrow B \to D$.
- $A \perp\!\!\!\perp C$? **Yes**: colliders at B ($A \leftrightarrow B \leftarrow C$) and at D ($A \leftrightarrow B \to D \leftrightarrow C$).
- $A \perp\!\!\!\perp C \mid D$? **No**: $B \in An(D)$ opens the collider at B.
- $D \perp\!\!\!\perp C \mid B$? **No**: $D \leftrightarrow C$ are adjacent.

1.2.7 Implementation: graph algorithms in Python

Motivation. Topological ordering and d-separation in code.

1.2.7.1 Topological order

A DAG has a source; removing it leaves a DAG. Recurse:

```
def topological_order(G):
    for V in G.Vs:                  # for every node
        if not G.Pa(V):             # if there is no parent for V
            return [V] + topological_order(G - V)
    return []
```

This is quadratic and silent on cycles. Kahn's algorithm tracks in-degrees and detects cycles:

```
def topological_order(G):
    order = deque()
    in_degrees = {V: len(G.Pa(V)) for V in G.Vs}
    while in_degrees:
        for v, v_deg in in_degrees.items():
            if v_deg == 0:
                break
        else:
            raise ValueError('cyclic')
        order.append(v)
        del in_degrees[v]
        for c in G.Ch(v):
            in_degrees[c] -= 1
    return order
```

The `for`–`else` branch runs iff no zero-in-degree node remains, i.e. iff there is a cycle.

1.2.7.2 d-separation via reachability (Markovian graphs)

X, Y are d-connected given $\mathbf{Z}$ iff some path has every triplet $A - B - C$ satisfying:

$$\begin{array}{ll} A \to B \to C & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \leftarrow C & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \to C & \text{only if } B \notin \mathbf{Z}, \\ A \to B \leftarrow C & \text{only if } B \in An(\mathbf{Z}). \end{array}$$

BFS/DFS over states (arrival direction, B): into B ($\to B$) or out of B ($\leftarrow B$).

```
def d_separation(G, X, Y, Zs):
    q = visited_deque({('->', X), ('<-', X)})
    while q:
        direction, B = q.pop()
        if B == Y:
            return False
        if direction == '->':          # arrived with arrowhead into B
            if B in G.An(Zs):          # collider: open iff B in An(Z)
                q.extend({('<-', C) for C in G.Pa(B)})
            if B not in Zs:            # chain continues
                q.extend({('->', C) for C in G.Ch(B)})
        else:                          # arrived via tail of B (B -> previous)
            if B not in Zs:
                q.extend({('->', C) for C in G.Ch(B)})
                q.extend({('<-', C) for C in G.Pa(B)})
    return True
```

`visited_deque` skips visited states, so each (direction, node) is processed once: linear time. X may be left in either direction.